

RAFFLES INSTITUTION
2018 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

9744/02

Paper 2 Structured Questions

13th September 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 12
2	/ 15
3	/ 17
4	/ 12
5	/ 12
6	/ 12
7	/ 10
8	/ 10
Total	/ 100

This document consists of **22** printed pages.



Raffles Institution
Internal Examination

Answer **all** questions.

- 1 Phosphofructokinase (PFK) is an allosteric enzyme made of 4 subunits and controlled by many activators and inhibitors which regulate glycolysis. PFK phosphorylates fructose-6-phosphate to form fructose-1,6-bisphosphate. This enables the cell to increase or decrease the rate of glycolysis in response to the cell's energy requirements. For example, a high ratio of ATP to ADP will inhibit PFK and glycolysis.

Fig 1.1 shows the effect of low ATP on phosphofructokinase (PFK) activity.

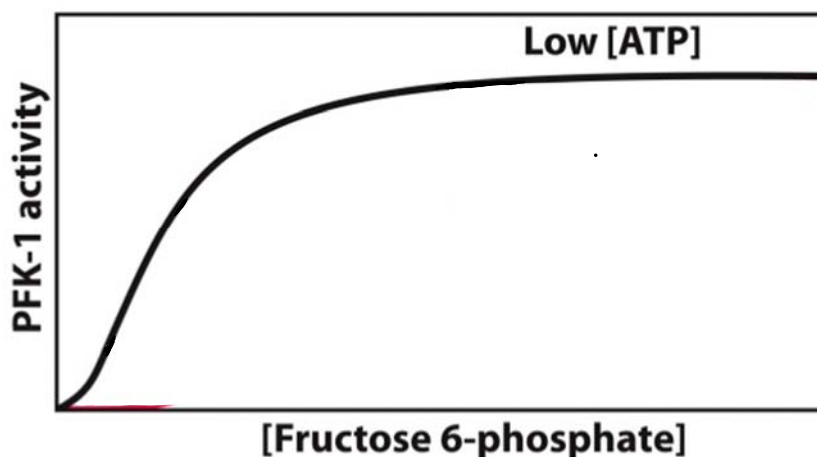


Fig. 1.1

- (a) (i) On Fig. 1.1, draw a graph to show the effect of high ATP on PFK activity. [1]
- (ii) Explain the graph you have drawn.

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- (iii) Name a molecule that will act as an allosteric activator of PFK.

..... [1]

- (b) Fig. 1.2 shows a PFK molecule.

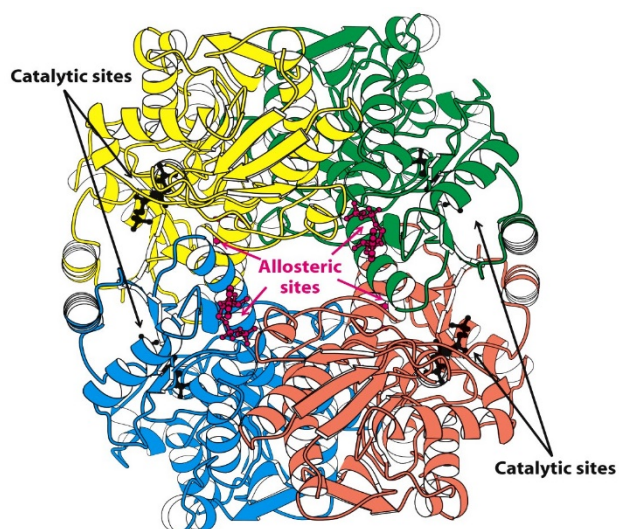


Fig. 1.2

- (i) Describe the structure of PFK.

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- (ii) Explain how a change in pH may affect PFK activity.

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..... [3]

[Total : 12]

- 2 A student wanted to investigate the difference between gene regulation of lactose metabolism in humans and bacteria.

The enzyme, lactase, is responsible for breaking down lactose in humans. *LCT* gene coding for lactase is found on chromosome 2, where its molecular nucleotide location is from base pairs 135,787,840 to 135,837,195.

Deletion constructs of the *LCT* gene were designed with various sequences on chromosome 2 deleted. These nucleotide positions are all upstream of the *LCT* gene, and were selected as they were found to be bound by molecules (A, B and C) which exert influence over *LCT* gene regulation.

Fig. 2.1 shows the map of a particular portion of chromosome 2, where relative positions of these sequences as well as the molecules associated with them are shown.

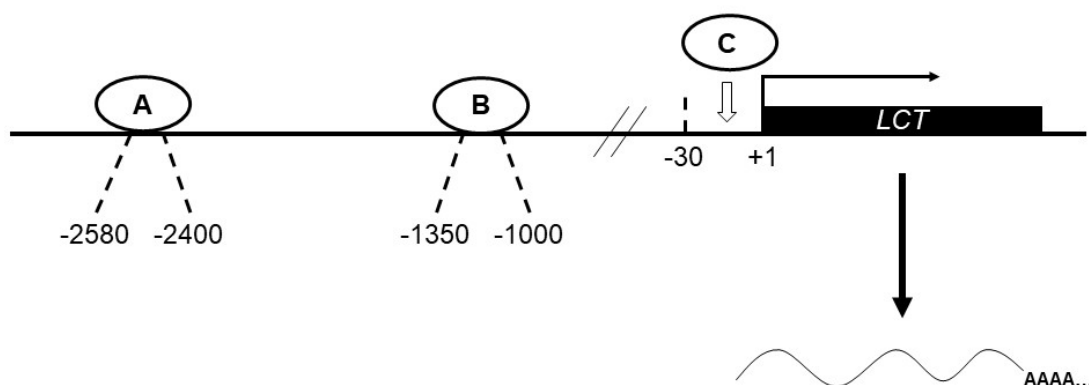


Fig 2.1

In order to quantify the effect of deletion on regulation of *LCT*, the student measured the amount of *LCT* mRNA produced, and the results are summarised in Table 2.1.

Deletion construct	Nucleotides deleted	Relative mRNA quantity / a.u.
Wild type	-	268
1	-30 to 0	0
2	-1350 to -1000	550
3	-2580 to -2400	76

Table 2.1

- (a) With reference to Fig 2.1 and Table 2.1,
- (i) identify the type of molecule as well as the type of regulatory sequences studied in the experiment.

	Type of molecule	Position	Type of Regulatory Sequence
A		-2580 to -2400	
B		-1350 to -1000	
C		-30 to 0	

[3]

- (ii) explain how you arrived at the identity of molecule **B** as well as the type of regulatory sequence at -1350 to -1000.

.....

 [2]

- (b) The *LCT* mature mRNA excluding its poly-A tail is much shorter than the *LCT* gene. Account for this difference.

.....

 [2]

- (c) The *lac* operon is responsible for lactose metabolism in bacteria.

With reference to the *lac* operon, describe one difference in the regulation of lactose metabolism in bacteria and humans.

..... [1]

- (d) The student wanted to investigate how glucose and lactose impact gene expression of *lac* operon and growth of *E. coli*. Fig. 2.2 shows the utilisation of glucose and lactose as carbon sources in a culture of bacteria.

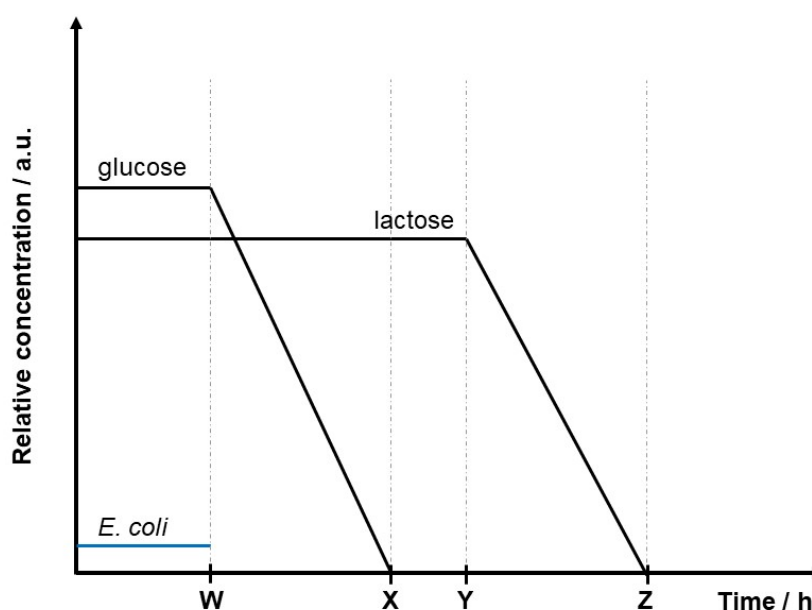


Fig 2.3

- (c) (i) Complete the graph by showing the concentration of bacteria in culture over time from **W** to **Z**. [1]

Using your knowledge of the *lac* operon, explain the shape of the graph for lactose from

- (ii) time **W** to **X**, and

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..... [2]

- (ii) Time **Y** to **Z**

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[Total : 15]

3 During meiosis, the cell divides to produce gametes.

(a) Discuss the role of centromere in the production of gametes.

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(b) The graph in Fig. 3.1 shows the length of the spindle fibres during meiosis.

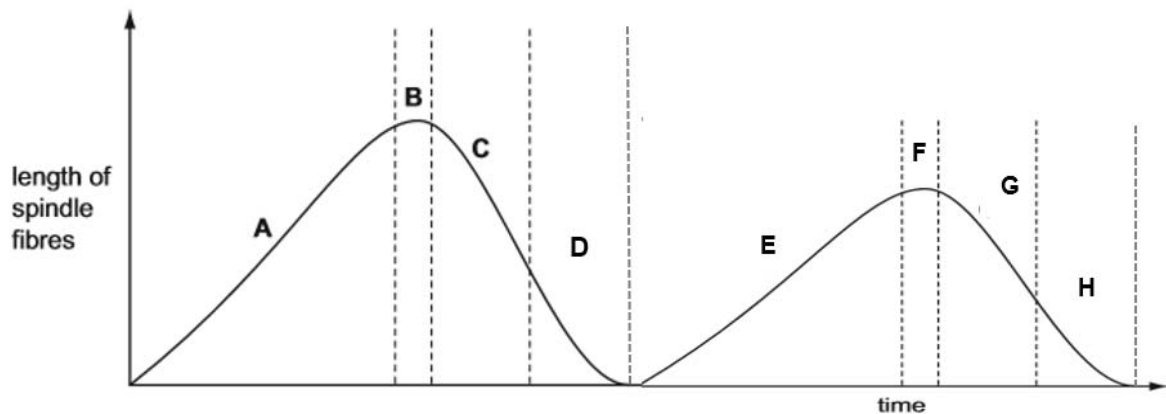


Fig. 3.1

(i) In which regions of the graph did the centromeres detach from the spindle fibres?

..... [1]

(ii) Fig. 3.2 shows a particular stage in meiosis

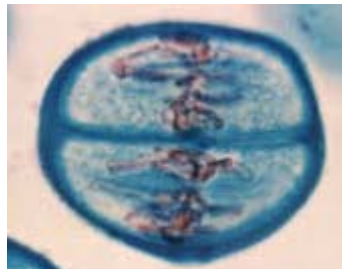


Fig 3.2

Which region on the graph corresponds to the stage shown in the Fig. 3.2.

..... [1]

- (ii) Name and identify 2 corresponding stages on the graph in Fig 3.1 that contributes to genetic variation in gametes, and explain how they bring about genetic variation.

Stage

Explanation

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Stage

Explanation

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..... [6]

- (c) Most human traits as well as medical conditions are under genetic influence.

A gene is found to control a rare disease, Wiskott–Aldrich syndrome (WAS) which is characterised by eczema, low platelet count and immune deficiency.

Another gene that controls hair type has 2 alleles. 1 allele results in straight hair, another codes for curly hair. Presence of both results in wavy hair.

In **couple 1**, a normal female with straight hair married a wavy-haired male suffering from WAS. Predicted phenotypic ratio of their offspring is as follows:

	WAS	Hair
Female	All normal	1 wavy : 1 straight
Male	All normal	1 wavy : 1 straight

For **couple 2**, a wavy-haired female suffering from WAS married a wavy-haired normal man. Predicted phenotypic ratio of their offspring is as follows:

	WAS	Hair
Female	All normal	1 curly : 2 wavy : 1 straight
male	All affected	1 curly : 2 wavy : 1 straight

- (i) What is the mode of inheritance of the WAS disease?

..... [1]

- (ii) Use suitable symbols to represent the alleles of the gene controlling the WAS disease.

..... [1]

(iii) Use a genetic diagram to explain the results of **couple 2**.

[4]

[Total : 17]

- 4 Fig. 4.1 shows the results of experiments investigating the effect of different light intensities on the rate of photosynthesis of cucumber plants measured as $\text{mm}^3 \text{CO}_2$ uptake per cm^2 leaf area per hour. The experiments were carried out at different temperatures and carbon dioxide concentrations.

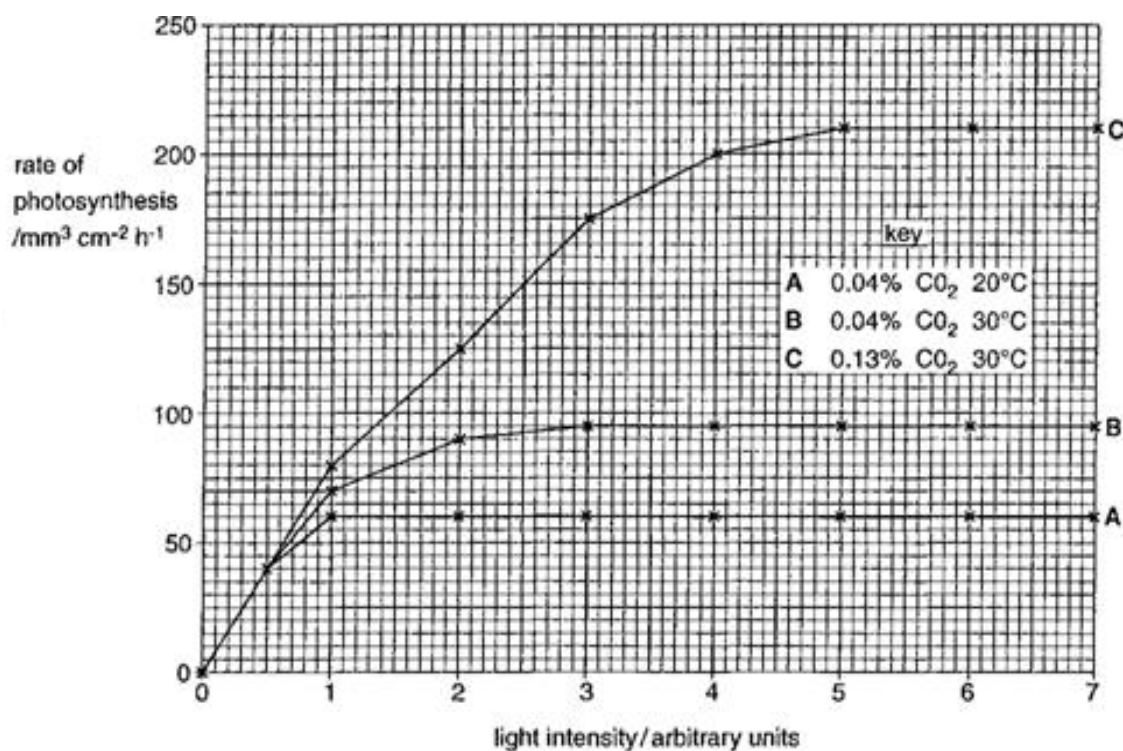


Fig. 4.1

- (a) (i) With reference to Fig. 4.1, state the best conditions for the growth of cucumber plant.

..... [1]

- (ii) Define the term limiting factor.

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..... [1]

- (iii) With reference to Fig. 4.1, explain reasons for the difference between curves B and C.

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- (b) Fig. 4.2 shows the absorption spectra of various leaf pigments and the action spectrum for photosynthesis.

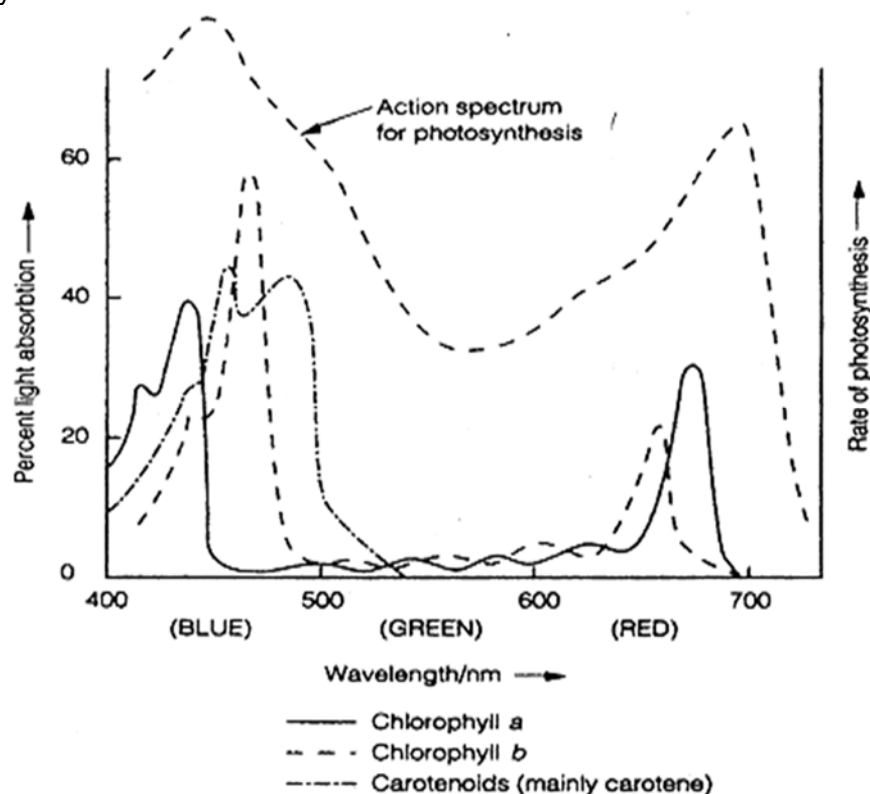


Fig. 4.2

With reference to Fig. 4.2,

- (i) explain why most plants are characteristically green, and

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..... [2]

- (ii) explain the shape of the action spectrum.

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- (c) In conducting experiments to obtain the action spectrum for photosynthesis, explain why the use of organic solvents such as acetone or ether is necessary to extract photosynthetic pigments.

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..... [2]

[Total: 12]

- 5 Using two mutant strains of *E. coli* (strain **A** and strain **B**) with different nutritional requirements, scientists showed that gene transfer could happen between bacteria. Strain **A** would grow only if the medium was supplemented with methionine and strain **B** would only grow if supplemented with threonine. When strains **A** and **B** are mixed together, some of the progeny were observed to regain the ability to grow without added nutrients in the medium.

Fig. 5.1 shows what was found when scientists observed the mixture of strains **A** and **B** under a microscope.

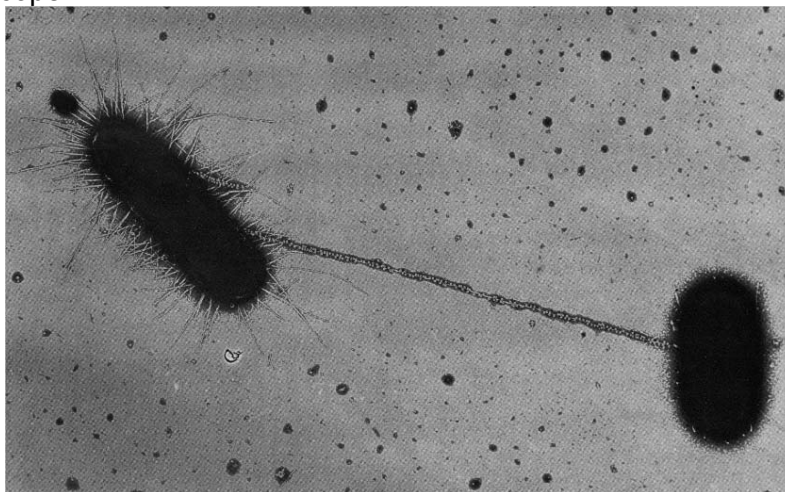


Fig. 5.1

- (a) Name and describe the method of gene transfer seen in Fig. 5.1.

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.....[4]

The following experimental set-up was used to investigate more about the method of gene transfer seen in Fig. 5.1. A suspension of strain **A** was placed in one arm of a U-tube and strain **B** was placed in the other arm. Liquid may be transferred between the arms but bacterial cells cannot pass through the center filter.

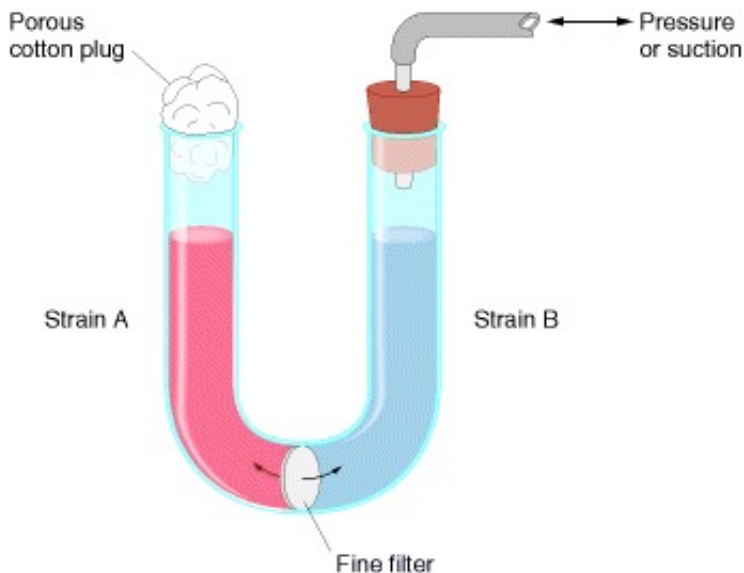


Fig. 5.2

- (b)** After several hours of incubation, the cells were transferred to minimal medium (without methionine and threonine). Predict and explain if any bacterial growth would be observed.

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[2]

- (c)** It was suggested that the scientists should set up a test-tube containing a mixture of strain **A**, strain **B** and an enzyme nuclease.

- (i)** Predict if the bacterial progeny from this tube can survive in minimal medium without the addition of methionine and threonine.

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 [2]

- (ii) State the purpose of setting up this tube.

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 [1]

- (d) Further investigation revealed that the genes controlling methionine synthesis and threonine synthesis are found on the bacterial chromosome. They can be transferred after the F plasmid integrates with the bacterial chromosome. The method of transfer is illustrated in Fig. 5.3.

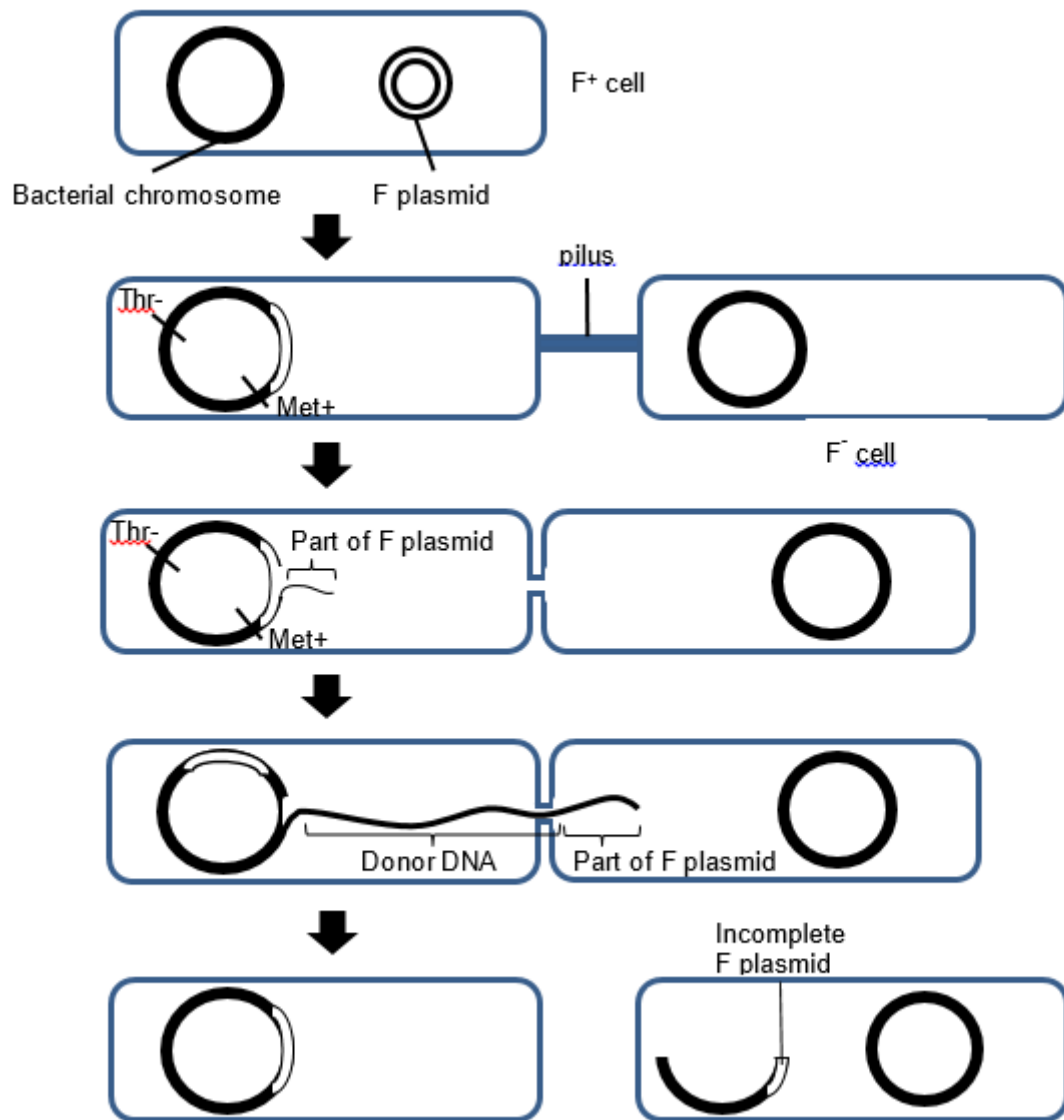


Fig. 5.3

- (i) With reference to Fig. 5.3, determine if strain **A** or strain **B** is the F^+ cell.

..... [1]

- (ii) In cases of gene transfer shown in Fig. 5.3, the F plasmid-bacterial chromosome is usually not transferred completely to the recipient cell. Random breakage interrupts the process. The recipient cell is unable to go on to donate any genes to other cells.

With reference to Fig. 5.3, suggest if the recipient cell is considered an F⁺ or an F⁻ cell after the transfer. Explain your answer.

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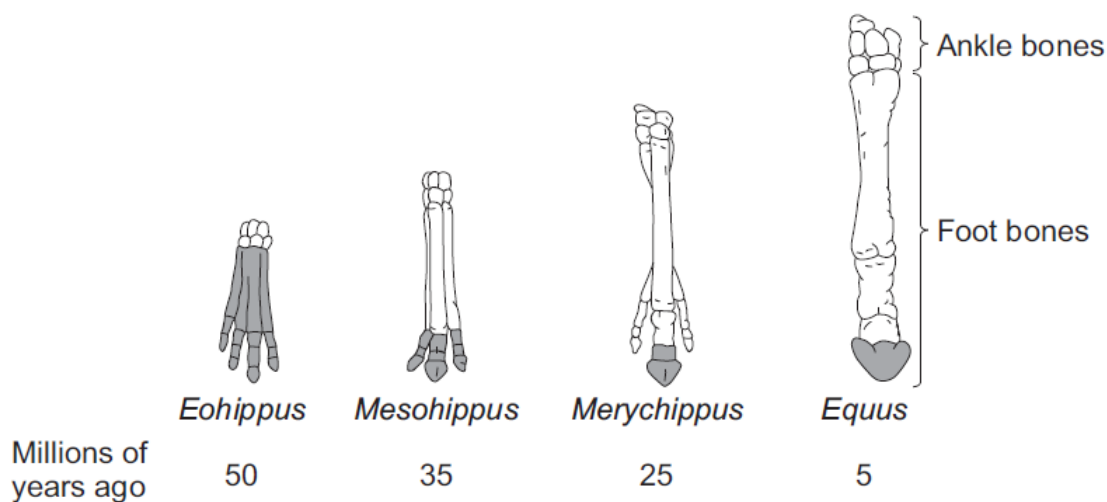
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[Total: 12]

- 6 Fig. 6.1 shows changes in the foot bones of four ancestors of modern horses over the past 50 million years.



Key: The shaded bones are the bones which touched the ground.

Fig. 6.1

- (a) Describe two changes in the foot bones of horses that have taken place over the past 50 million years.

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..... [2]

- (b) *Eohippus* lived in swampy areas with soft mud.
Since then, the ground in the habitat has become drier and harder.
All of the horse ancestors were preyed upon by other animals.

(i) Explain **one** advantage to *Eohippus* of the arrangement of bones in its feet.

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..... [1]

(ii) Based on the information given, explain how the changes in the arrangement of the foot bones of horses support Darwin's theory of evolution by natural selection.

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..... [4]

- (c) (i) Fossils of *Eohippus*, *Mesohippus* and *Merychippus* have been unearthed only in the North American continent. Explain how the biogeographical distribution of these fossils support Darwin's theory of evolution.

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..... [3]

- (ii) The first “modern” one toed horses *Equus* also arose in North America, around 5 million years ago, but quickly spread into the Asia and Europe, when the land bridge formed between continental America and Asia. They then diversified into the modern species we know today - horses, donkeys, asses and zebras.

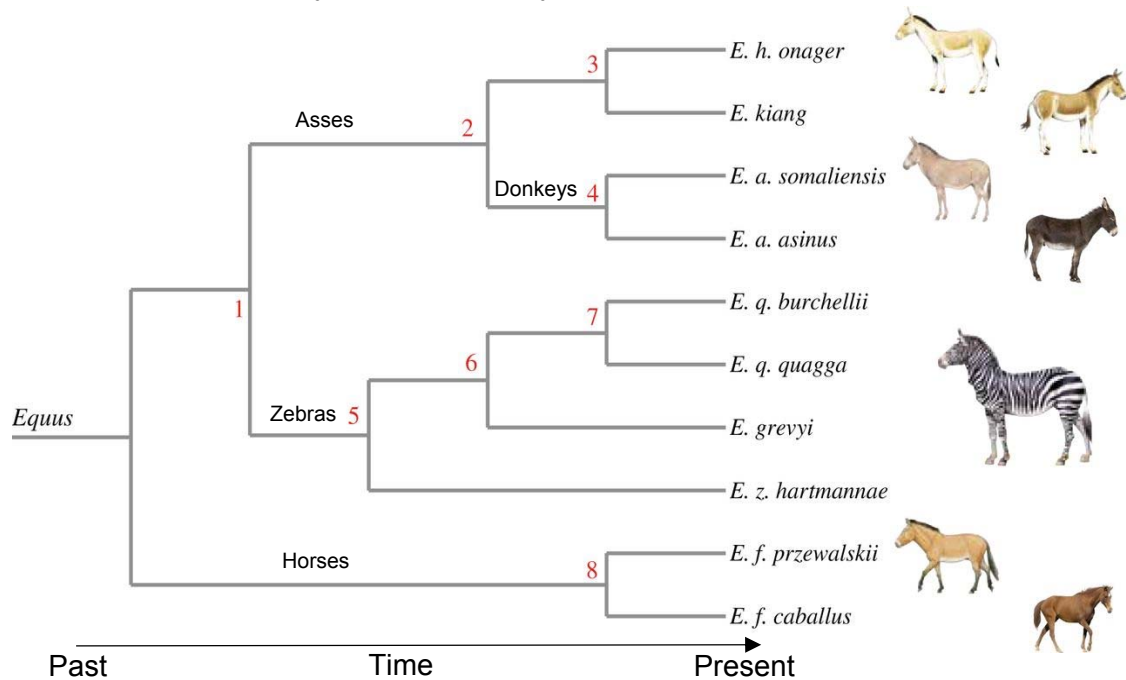


Fig. 6.2

Explain how scientists created the phylogram shown in Fig. 6.2.

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..... [2]

[Total :12]

- 7 The immune system protects the human body against pathogen's infection of the host and subsequent development of disease. Fig. 7.1 shows the action of a macrophage on a pathogenic bacteria.

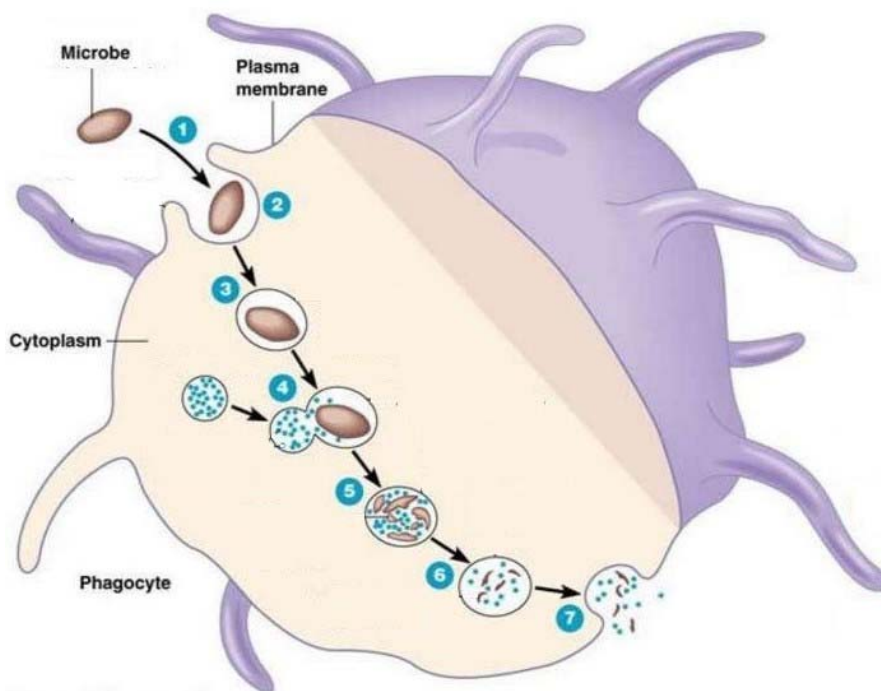


Fig. 7.1

- (a) With reference to Fig. 7.1, identify the steps and explain the processes that kills the pathogenic bacteria.

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.....[2]

- (b) Describe how the humoral immune response enhances the action of macrophages on the pathogenic bacteria.

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.....[3]

- (c) Tuberculosis (TB) is a highly infectious air-borne disease that affects the lungs, brain, lymph nodes, kidneys and bones. In the recent years, there is a rise in the number of TB cases. In the year 2017, 1,536 new cases of TB infections were reported.

TB is caused by the bacteria, *Mycobacterium tuberculosis*. The process shown in Fig. 7.1 has limited efficacy in clearing *M. tuberculosis* that infected the hosts.

Explain how *M. tuberculosis* protects itself from the host immune system.

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[2]

- (d) Patients diagnosed with TB are treated with a regime that uses a “cocktail” of drugs which include isoniazid, rifampin, pyrazinamide and ethambutol.

Fig 7.2 show the mechanism of action of rifampin on RNA polymerase.

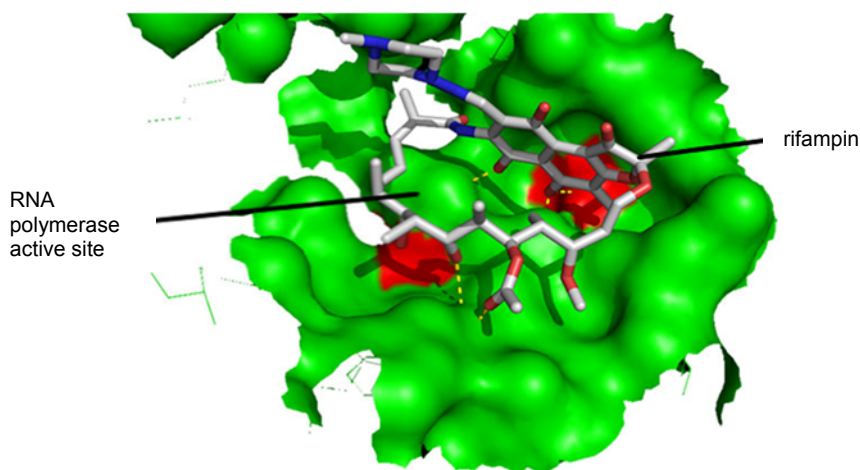


Fig. 7.2

With reference to Fig. 7.2, explain how rifampin acts to achieve bactericidal effects on *M. tuberculosis*.

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[3]

[Total:10]

- 8 Sea ice is an integral part of the Arctic Ocean. The extent of area covered by Arctic sea ice is an important indicator of changes in global climate because warmer air and water temperatures are reducing the amount of sea ice present.

Fig. 8.1 shows Arctic sea ice extent for the months of September and March of each year from 1979 through 2016. September and March are when the minimum and maximum extent typically occur each year.

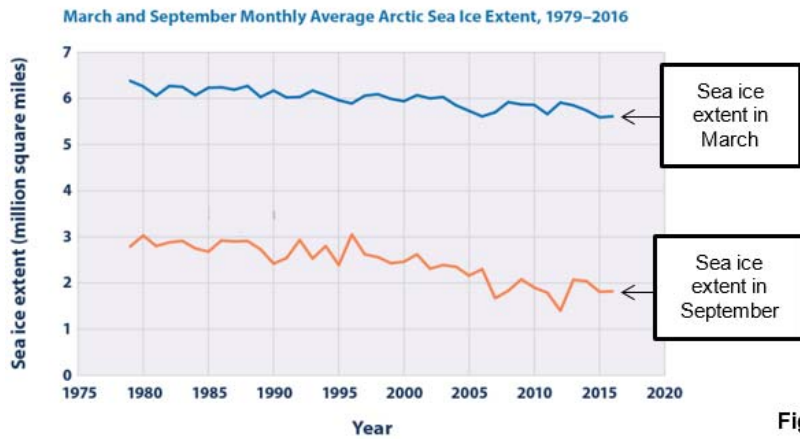


Fig. 8.1

Fig. 8.2 shows the start and end of the sea ice melt season.

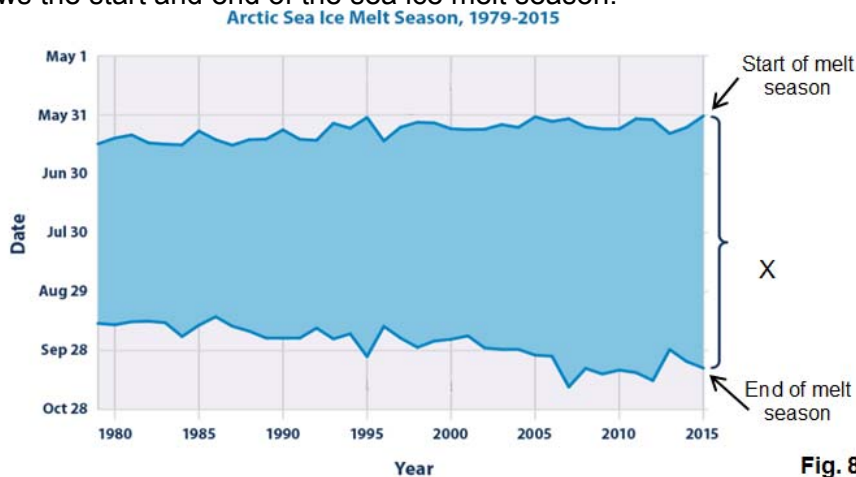


Fig. 8.2

Fig. 8.3 shows the distribution of Arctic sea ice extent by age group during the week in September with the smallest extent of ice for each year.

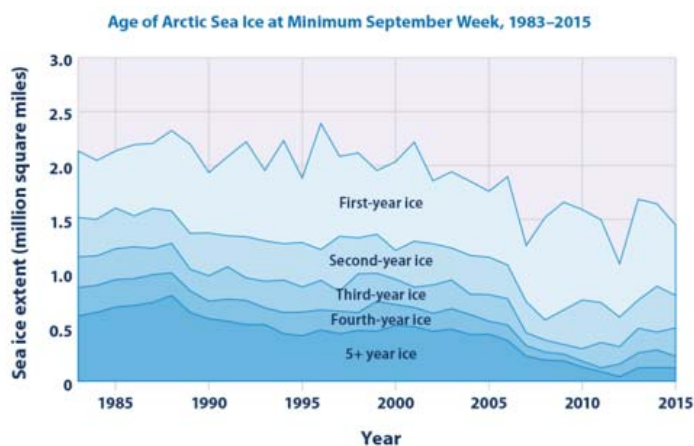


Fig. 8.3

- (a) With reference to Fig. 8.1, suggest why the sea ice extent in March is different from that in September.

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.....[1]

- (b) With reference to Fig. 8.2,
(i) identify what **X** denotes.

..... [1]

- (ii) describe how **X** changes from 1979 to 2015.

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..... [1]

- (ii) Explain how this is linked to the changes in sea ice extent as shown in Fig. 8.1 and Fig. 8.3.

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..... [3]

- (c) A number of animals living in the Arctic are able to change their coat colour as the seasons change.

The ability to change coat colour is a result of their ability to control pigment production in response to variation in lengths of day or night.

These animals are able to detect the lengths of day or night via a light-sensitive receptor known as melanopsin that is found in their retina. Fig. 8.4 shows the structure of melanopsin.

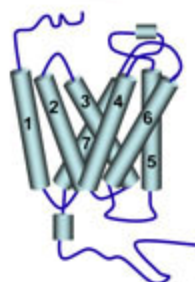


Fig. 8.4

- (i) State the class of receptors that melanopsin belongs to.

..... [1]

Melanopsin is known to activate an enzyme known as phospholipase C via the signaling pathway shown in Fig. 8.5.

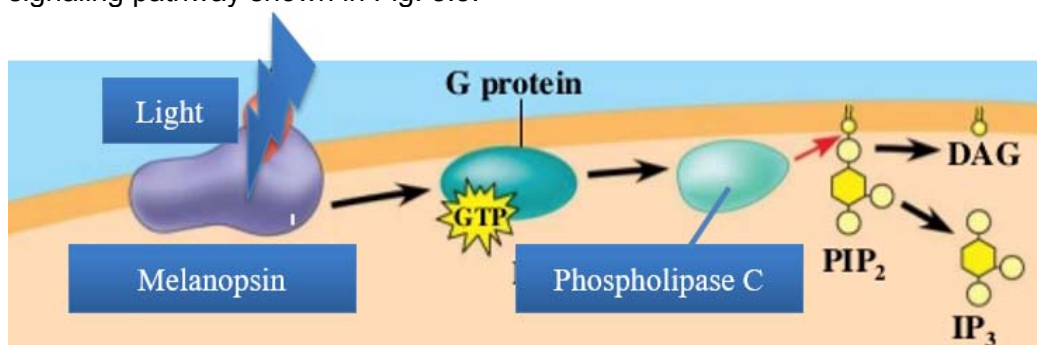


Fig. 8.5

- (ii) With reference to Fig. 8.5, describe how phospholipase C could become activated.

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..... [3]

[Total : 10]

End of Paper

